

Autodesk® Scaleform®

字體和文字配置

APIs

Scaleform 4.3

Scaleform C++

Maxim Shemanarev, Michael Antonov

2.2

2012 6 21

Autodesk Scaleform Corporation
6305 Ivy Lane, Suite 310
Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

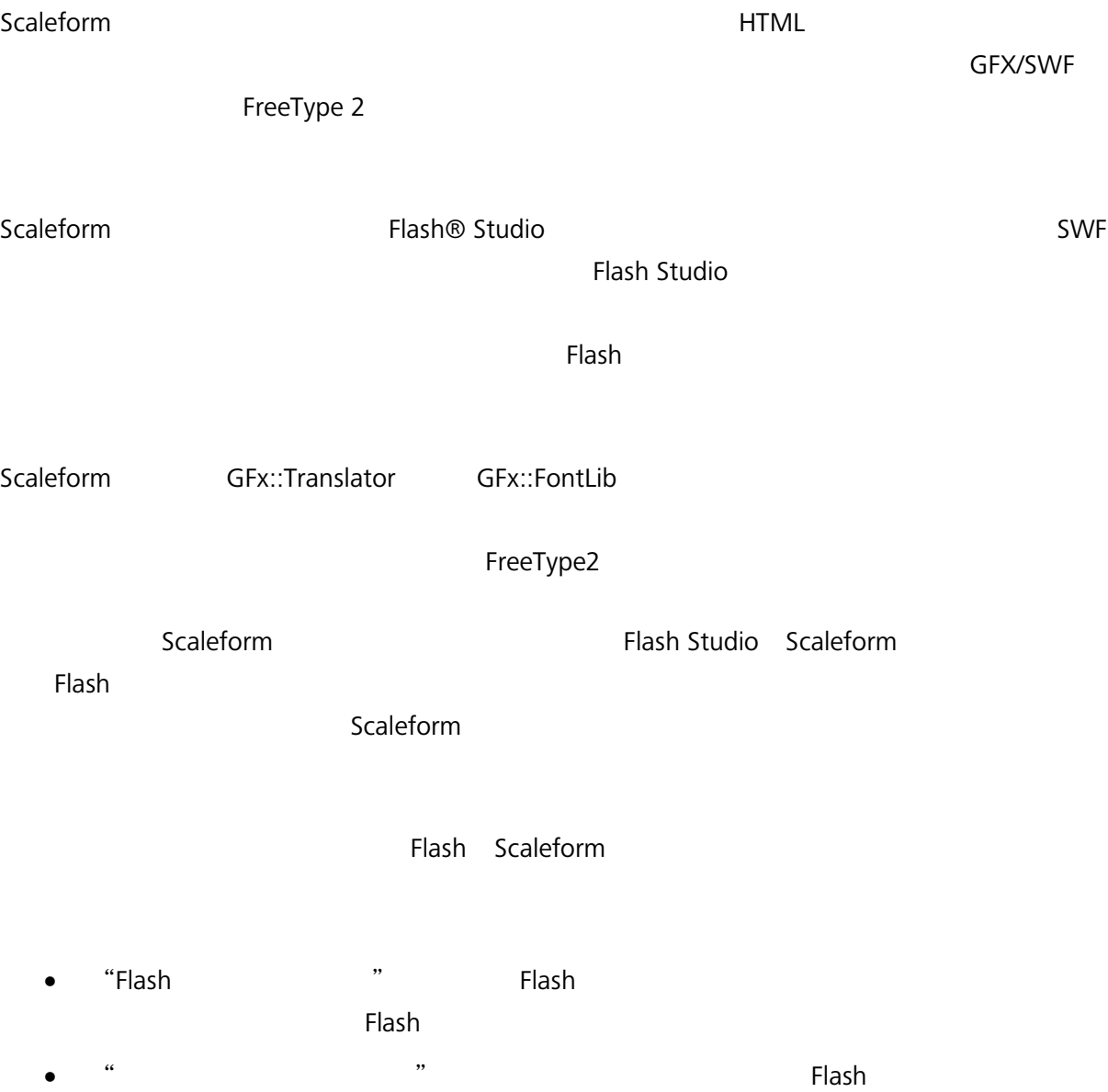
(301) 446-3200

(301) 446-3199

1	引言: Scaleform 中的字體	1
1.1	Flash	1
1.1.1		2
1.1.2		3
1.1.3		4
1.1.4		5
1.2		6
2	1	7
2.1		7
2.1.1		8
2.1.2		10
2.2	gfxfontlib.swf	10
3	2	12
3.1		12
3.1.1		14
3.1.2	Scaleform Player	15
3.1.3	設備字體仿真	16
3.1.4		17
3.2		19
4	3	20
4.1		20
4.2	Gfx::FontMap	22
4.3	Gfx::FontLib	23
4.4	Gfx::FontProviderWin32	23
4.4.1		24
4.4.2		25
4.5	Gfx::FontProviderFT2	26

4.5.1	FreeType	27
4.5.2	將字體映射到記憶體	27
5	4	29
5.1	配置字形緩存	29
5.2		31
5.3	- gfxexport	33
5.4	- gfxexport	33
5.5		35
5.6		36
6	5	39
6.1		39
6.2		40
6.3		41
6.4	ActionScript	42

1 引言：Scaleform 中的字體



1.1 Flash

Flash Studio

Flash

Studio

Flash

Scaleform

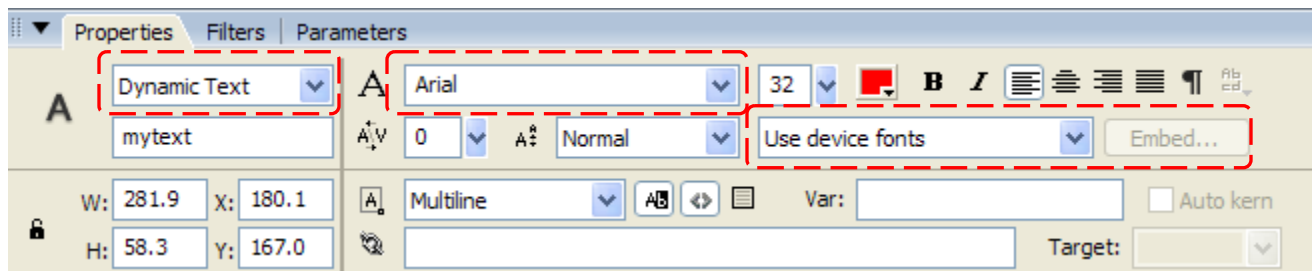
1 ---

1.1.1

Flash

Text Tool

Flash Studio



-
-
-

GFx::Translator

(a)

“Arial”

(b)

-
-
-
-

“Use device fonts”

“ ”

SWF
Flash

/

Scaleform

“ ” “ ”

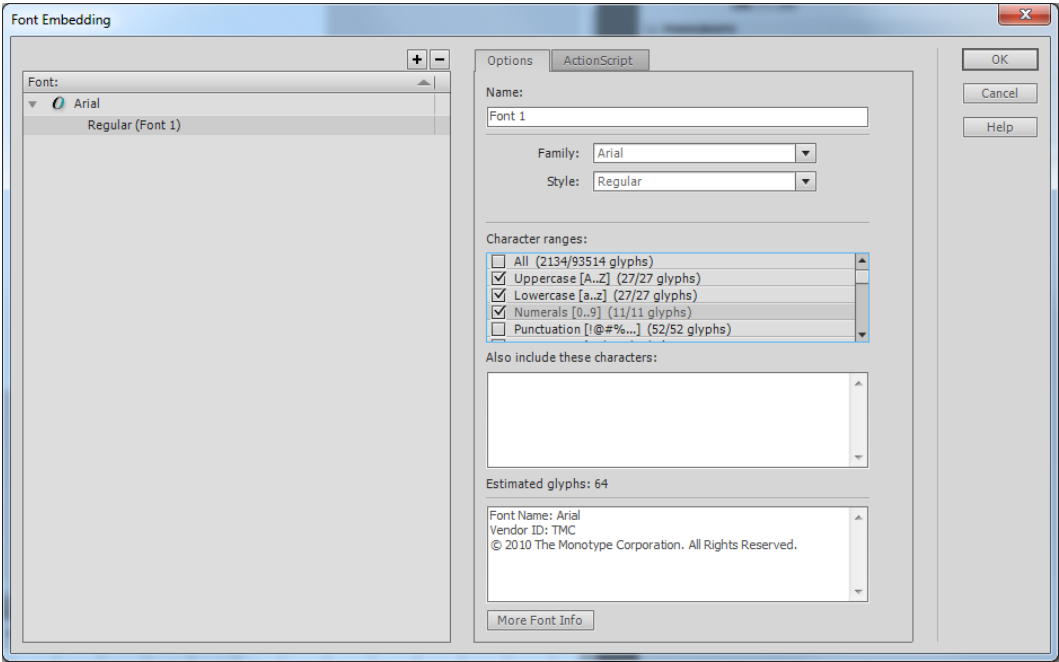
1.1.2

, TextField() (

,)

Scaleform Scaleform

Embed



) , (Arial Arial Style (

Flash Studio

SWF/GFx

Scaleform

Xbox 360 PS3

Wii

Windows Mac Linux

1.1.3

Scaleform ,

	SWF	Scaleform
1	1 KB	450 K

114	–	+	12 KB	480 K
596	–		70 KB	630 K
7,583	–		2,089 KB	3,500 K
18,437	–		5,131 KB	8,000 K

“Arial Unicode MS”

500

150K

1.1.4

1.
- Flash
2.
- Scaleform

/

SWF/GFx

Scaleform
3.
- IME
4.
- GfxEport

(-fc)

,

10%-30%,

(5.3)

Scaleform

IME

1.2

Scaleform

- 1 _____
- 2 _____
- 3 _____
- 4 - - - -

Scaleform

Scaleform

2

- Scaleform
-
-

Flash

SWF/GFx

Scaleform
3 “ ”

Scaleform

Scaleform
FreeType

2 Scaleform

Scaleform
gfxexport

2 1

1.

2.

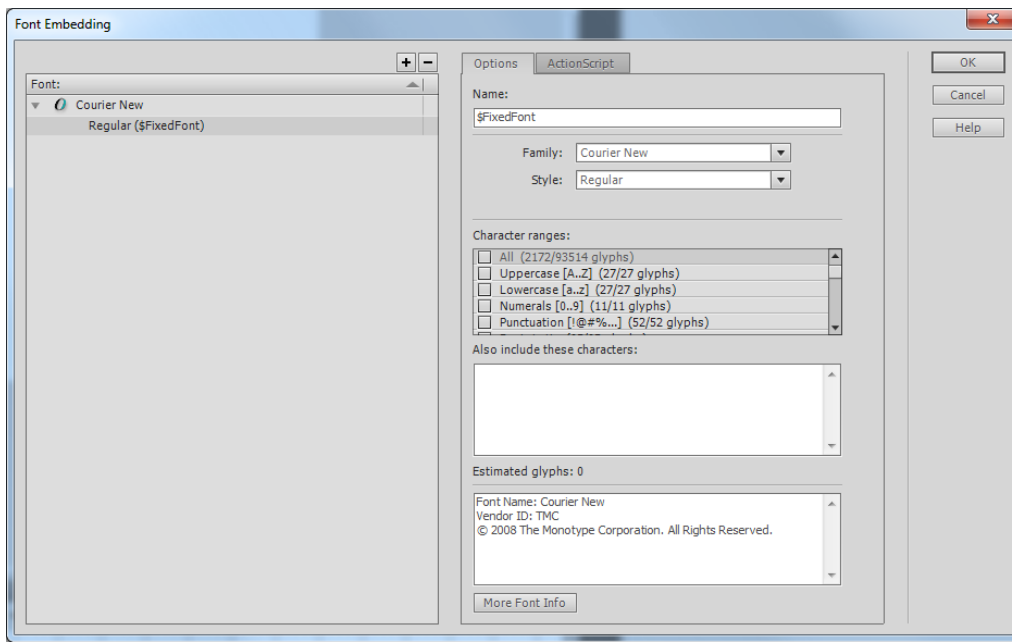
3.

Scaleform ,
, (: fonts_en.swf fonts_kr.swf)
Flash SWF “fonts_en.swf
“ Flash

2.1

Flash Studio

“New Font...”



FLA

“\$FixedFont”

“\$ArialGame*”

“\$”

“ ”

“Font rendering method”

“

”

Flash

Studio

2.1.1

FLA

FLA

FLA

/

“Copy”

“Paste”

FLA

Flash

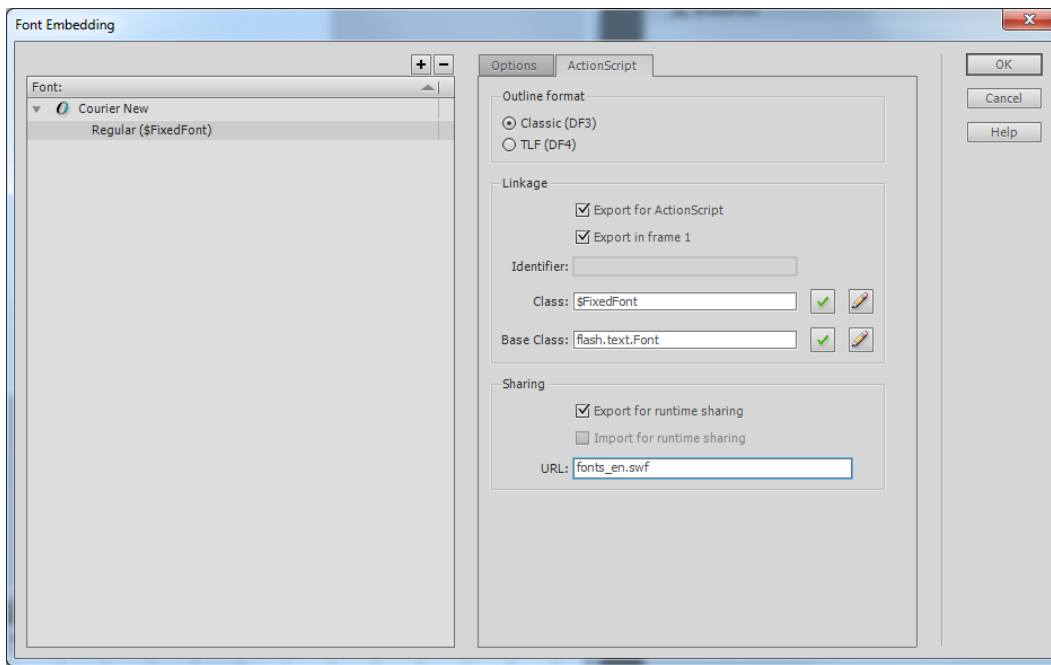
/

Properties (),

ActionScript

,

,



OK () ,Flash

Export for ActionScript (ActionScript) Export for runtime sharing ()
 Export in first frame ()
 URL SWF ,
 ; Gfx::FontLib , URL
 fonts_en.swf , , Gfx

```
Loader.SetDefaultFontLibName("fonts_en.swf");
```

```
fonconfig.txt GfxPlayer  
fontlib "fonts_en.swf"
```

```
fontlib
```

"Export: \$identifier"

/

SWF

Scaleform

SWF

TextField.htmlText
“Arial”

TextFormat

2.1.2

CS4 Flash Studio

```
        Windows          \          \          “          Unicode
        ”                “      ”          243
        11920            ,          ,      ,          ‘gfxfontlib.swf’

        ,      Flash CS5      ,          (Glyph),          ‘gfxfontlib.swf’
        (      ‘fonts_en.swf’      ‘fonts_jp.swf’      ),

        ,          CS4 Flash Studio(          ),          ‘gfxfontlib.swf’      (
        )          ,          GfX 4.0
```

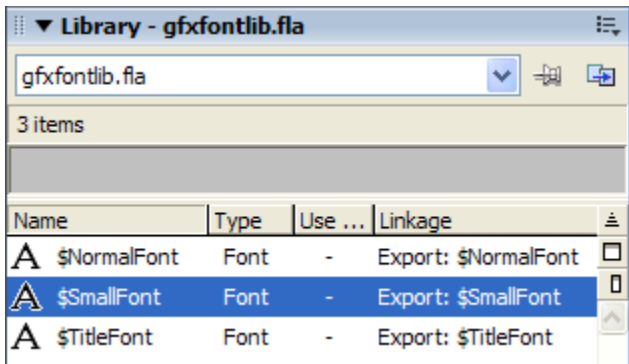
2.2 *gfxfontlib.swf*

```
        ,          ,

        ,          Flash Studio          FLA,
```

1. ‘\$TitleFont’ ‘\$NormalFont’
2. FLA
3. 1
- 4.
- 5.
6. fonts_<lang>.swf, ‘lang’ (:‘en’ – ‘ru’ – ‘jp’ –
‘cn’ – ‘kr’ –) , FontMap
</lang>

FLA

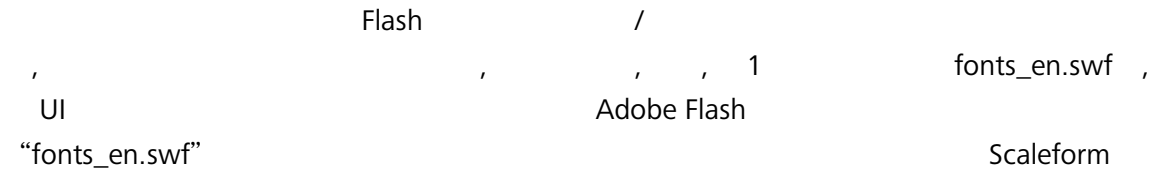


3 2

- 1.
- 2.
- 3.



3.1



```

, Loader::SetDefaultFontLibName (const char* filename) C++ , (
, !)'filename' , ,

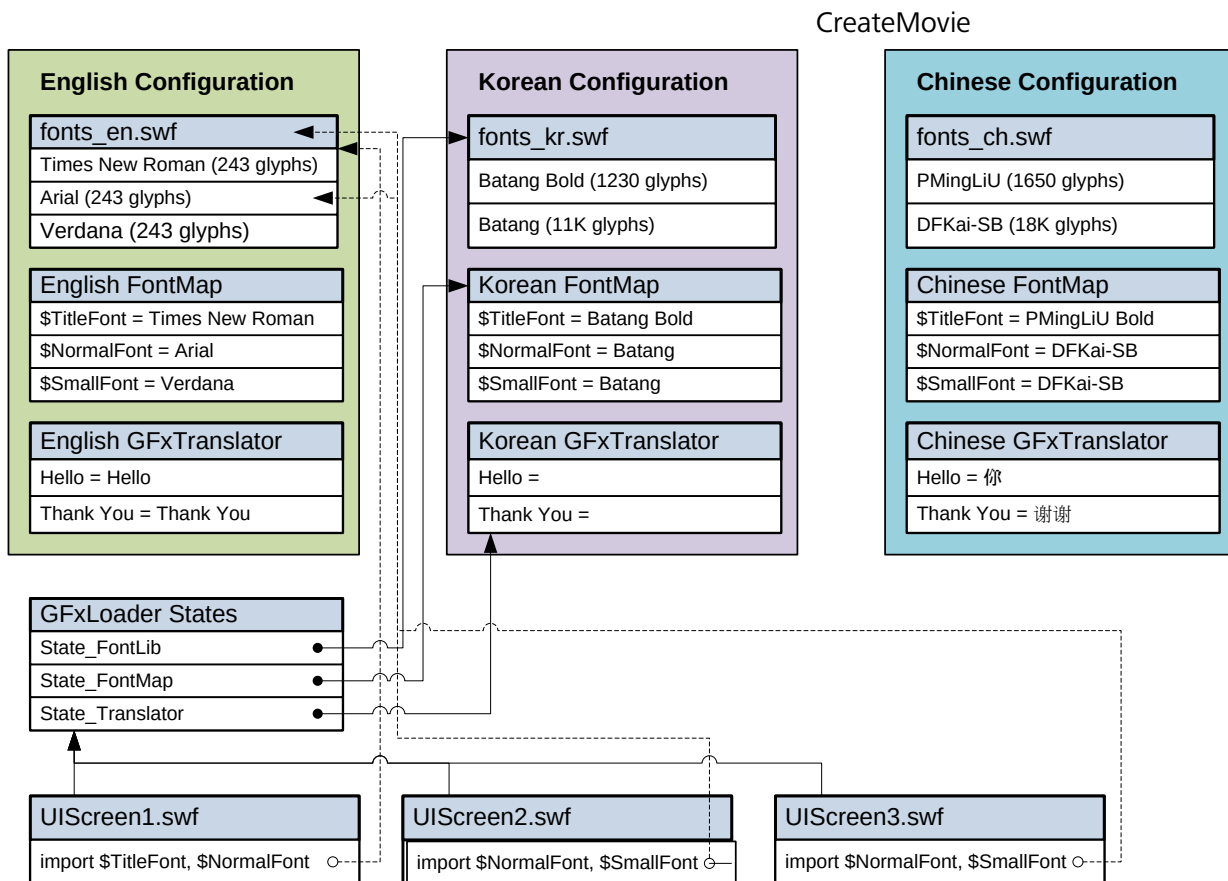
```

```
Loader.SetDefaultFontLibName("fonts_kr.swf");
```

SWF CreateInstance

- 1.
- 2.
3. "\$NormalFont"

Scaleform Gfx::Translator Gfx::FontLib Gfx::FontMap
Gfx::Loader Gfx::Loader::CreateMovie



			“UIScreen1.swf”
“UIScreen3.swf”	“fonts_en.swf”		
Scaleform		Scaleform	Gfx::FontLib
Gfx::FontMap states		:(1)	
Loader::SetDefaultFontLibName()		, (2)	UI
Gfx::FontLib			
,			
	“fonts_kr.swf”		

UIScreen

3.1.1

		SWF	Scaleform
			Gfx::FontMap Gfx::Translator
Scaleform		“fontconfig.txt”	

1.	1	(fonts_en.swf),	
2.			Gfx::Translator
3.		Gfx::FontMap	
4.		, fonts_kr.swf	

1.		Gfx::Loader	Gfx::Translator
2.	Gfx::FontMap	Gfx::Loader	MapFont
	3	Gfx::FontMap	

```

3.                                     ,      ,Gfx::FontProvider /
   Gfx::FontPackParams               ,      GlyphCacheConfig
4.           Gfx::FontLib             Gfx::Loader
5.           (swf                      ),      SetDefaultFontLibName(filename)

   Loader.SetDefaultFontLibName("fonts_en.swf");
           Gfx::FontLib               SWF/GFx
Gfx::FontLib::AddFromFile

6.           Gfx::Loader::CreateMovie

```

3.1.2 Scaleform Player

```

Scaleform Player                                     "fontconfig.txt"
SWF/GFX                                             'fontconfig.txt'
   /fc                                             Ctrl+N                                     F2
           HUD
           8           ASCII           UTF-16

```

```

[FontConfig "English"]
fontlib "fonts_en.swf"
map "$TitleFont" = "Times New Roman" Normal
map "$NormalFont" = "Arial " Normal
map "$SmallFont" = "Verdana" Normal

```

```

[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$TitleFont" = "Batang" Bold
map "$NormalFont" = "Batang" Normal
map "$SmallFont" = "Batang" Normal
tr "Hello" = "      "
tr "Thank You" = "      "

```

```

[FontConfig "Chinese"]
fontlib "fonts_ch.swf"
map "$TitleFont" = "PMingLiU" Bold
map "$NormalFont" = "DFKai-SB" Normal
map "$SmallFont" = "DFKai-SB" Normal
tr "Hello" = "      "
tr "Thank You" = "      "

```

[FontConfig "name"]

Scaleform Player HUD

fontlib "fontfile"	SWF/GFx GFx::FontLib fontlib
map "\$UIFont" = "PMingLiU"	GFx::FontMap GFx::FontMap \$UIFont default "fonts_en.swf" UI
tr "Hello" = " "	

Scaleform

Scaleform

XML

FontConfigParser.h/.cpp

Bin\Data\AS2\Samples\FontConfig and Bin\Data\AS3\Samples\FontConfig

Scaleform

"sample.swf"

"fontconfig.txt"

"sample.swf"

Scaleform Player

"fontconfig.txt"

3.1.3 設備字體仿真

GFx::FontMap

"Arial" "Verdana"

Scaleform Player

"Arial"

GFx::FontLib

"Batang"

"Batang"

"font_kr.swf"

GFx::FontLib

GFx::FontProviderWin32

GFx::FontProviderFT2

•

Flash

Adobe Flash Player

Scaleform

Flash

- “Anti-alias for animation”
TextField.antiAliasType Scaleform Player ActionScript
- Scaleform Adobe Flash
(
fonts_en.swf)
- ActionScript TextFormat Flash
“Arial” “\$TitleFont”
“fonts_en.swf”
“fonts_en.swf”

3.1.4

SWF , SWF , fontconfig.txt

1. FLA, ActionScript 2.0 3.0 ‘main-app.flasw’
2. FLA, ActionScript ‘fonts_en.swf’
3. ‘Library’() , , ‘New Font...’(...) \$Font1,
Family () Arial , Style () Regular
4. Character ranges () , English(), ‘Basic Latin’
5. ‘ActionScript’ ‘Export for ActionScript’(ActionScript) ‘Export for
frame 1’(1) ‘Export for run-time sharing’() ‘URL’ ,
‘fonts_en.swf’ ‘OK’() ActionScript 3.0,Flash Studio
‘A definition for this class could not be found...’(...) – ,
‘OK’()
6. 3 – 5, ‘\$Font2’, ‘Family’() / ‘Style’()
, ‘Family’() ‘Style’() , Flash Studio
, ‘Times New Roman’
7. FLA, ActionScript ‘fonts_kr.flasw’ ‘fonts_ru.flasw’,
, Korean(), ‘fonts_kr.flasw’

8. 'Library' () , , 'New Font...' (...) \$Font1, Family () (바탕), Style () Regular
9. Character ranges () , , 'Korean Hangul (All)' (())
10. 'ActionScript' 'Export for ActionScript' (ActionScript) 'Export for frame 1' (1) 'Export for run-time sharing' () 'URL' , 'fonts_en.swf' 'OK' () ActionScript 3.0, Flash Studio 'A definition for this class could not be found...' (...) - , 'OK' ()
11. 8 - 10, '\$Font2', 'Family' () , , '바탕체'
12. 'fonts_en.swf' 'fonts_kr.swf'
13. main-app.flas ('fonts_en.swf'), Library (), '\$Font1' '\$Font2', (Ctrl-C -> 'Copy' ())
14. 'main-app.flas', Library () , (Ctrl-V -> 'Paste' ()) , 'Import:' (:)
15. 'main-app.flas' 'Classic Text' () 'Dynamic Text' () 'Family' () '\$Font1*' '\$TEXT1', (, ID)
16. , '\$Font2*', '\$TEXT2'
17. 'main-app.swf',
18. , swf fontconfig.txt 'Notepad' ()
Unicode UTF-8

```
[FontConfig "English"]
fontlib "fonts_en.swf"
map "$Font1" = "Arial"
map "$Font2" = "Times New Roman"
tr "$TEXT1" = "This is"
tr "$TEXT2" = "ENGLISH!"
```

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$Font1" = "Batang"
map "$Font2" = "BatangChe"
tr "$TEXT1" = "이것"
tr "$TEXT2" = "은 한국이다!"
```

:	:	(바탕 바탕체),	Flash	SWF
(Batang BatangChe)	:	:	fontconfig.txt	

```

SWF
*.lst </swfname> , gfxexport -fntlst<swfname>

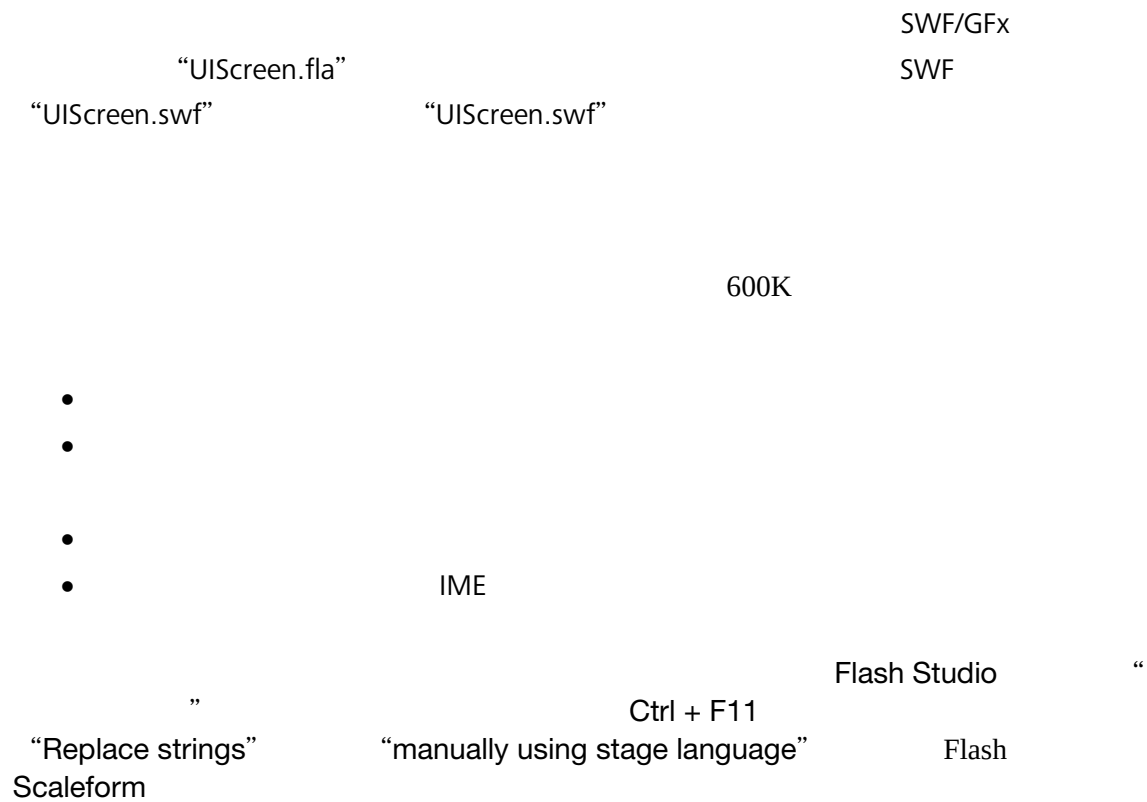
```

```

19. ( Unicode UTF-8!) ! GfxPlayer 'main-app.swf',
    'This is'( ) 'English!'( !) Ctrl-N Korean( ),
    fonts_kr.swf ( fonts_kr.swf )

```

3.2



4 3

Flash

HTML

1.

2. SWF / GfX

3. GfX::FontLib

SWF/GfX

4. GfX::FontProviderWin32

Flash

GfX::FontLib GfX::FontMap GfX::FontProvider
GfX::FontLib GfX::FontMap SWF/GfX

GfX::Loader::SetFontProvider

SWF

SWF

Scaleform

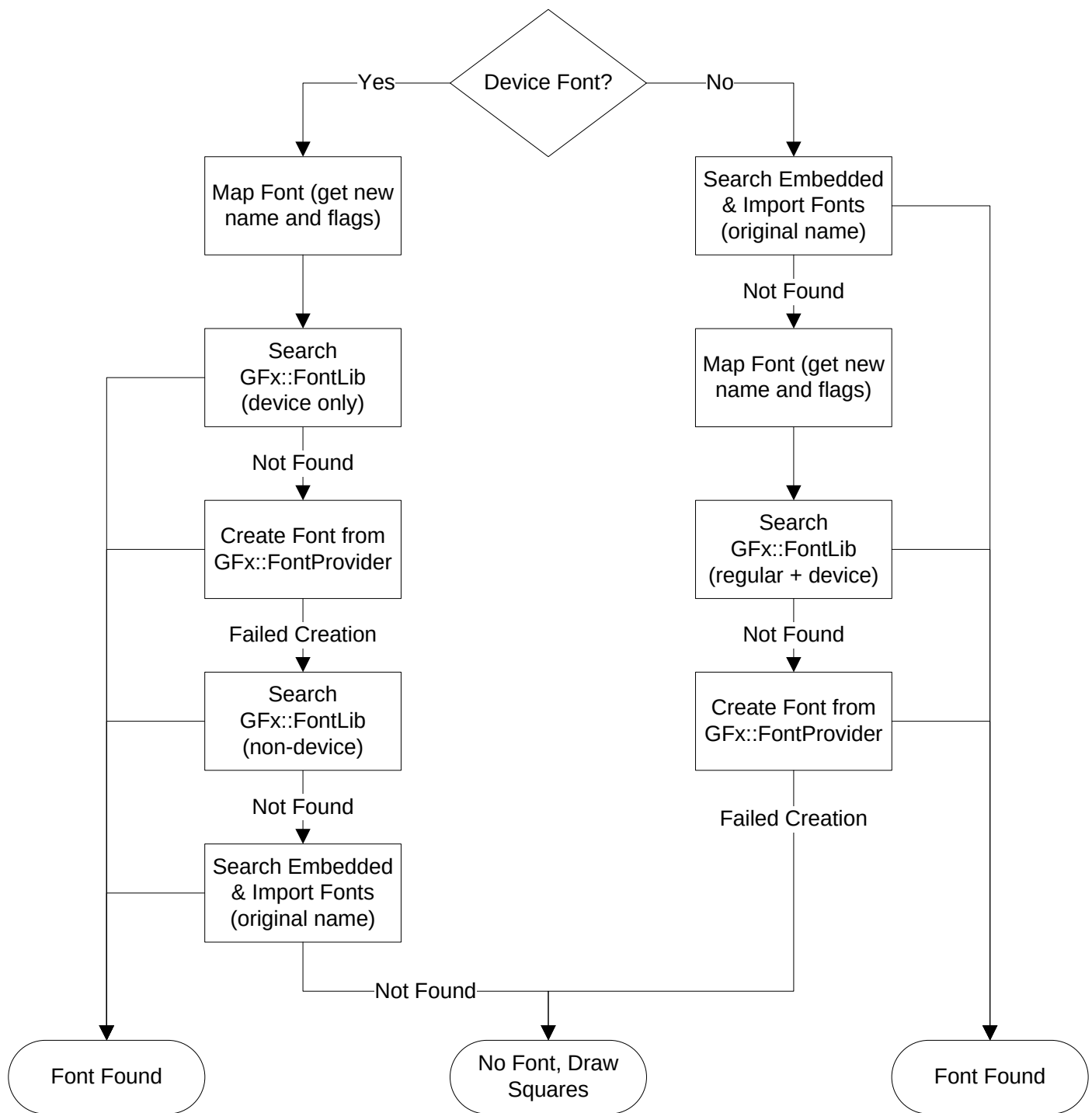
- GfX::FontProviderWin32 – Win32 APIs
- GfX::FontProviderFT2 – David Turner FreeType-2

4.1

Scaleform
“Use device fonts”

Adobe Flash

Scaleform



Gfx::FontLib Gfx::FontProvider

4.2 Gfx::FontMap

Gfx::FontMap

Gfx::FontMap

3

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$TitleFont" = "Batang" Bold
map "$NormalFont" = "Batang" Normal
map "$SmallFont" = "Batang" Normal
```

"\$TitleFont

C++

```
#include "Gfx/Gfx_FontLib.h"
. . .
Ptr<FontMap> pfontMap = *new FontMap;
Loader.SetFontMap(pfontMap);

pfontMap->MapFont("$TitleFont", "Batang", FontMap::MFF_Bold scaleFactor = 1.0f);
pfontMap->MapFont("$NormalFont", "Batang", FontMap::MFF_Normal scaleFactor = 1.0f);
pfontMap->MapFont("$SmallFont", "Batang", FontMap::MFF_Normal scaleFactor = 1.0f);
```

```
;      "$NormalFont" "$SmallFont"
      MapFont
      MFF_Original
      scaleFactor
```

scaleFactor 1.0f

Gfx::FontMap

Gfx::FontMap

Gfx::Loader::CreateMovie

Gfx::MovieDef

4.3 Gfx::FontLib

3 ,GFx::FontLib , (1) fonts_en.swf
 , (2)

GfX::FontLib

```
#include "Gfx/Gfx_FontLib.h"
. . .
Ptr<FontLib> fontLib = *new FontLib;
Loader.SetFontLib(fontLib);
fontLib->SetSubstitute("<default_font_lib_swf_or_gfx_file>");

Ptr<MovieDef> m1 = *Loader.CreateMovie("<swf_or_gfx_file1>");
Ptr<MovieDef> m2 = *Loader.CreateMovie("<swf_or_gfx_file2>");
. . .
fontLib->AddFontsFrom(m1, true);
fontLib->AddFontsFrom(m2, true);
```

AddFontsFrom

AddRef

M1 M2

/

Gfx::FontLib

4.4 Gfx::FontProviderWin32

Gfx::FontProviderWin32

Win32 API

GetGlyphOutline

```
#include "GfX/GfX_FontProviderWin32.h"
. . .
Ptr<FontProviderWin32> fontProvider = *new FontProviderWin32(::GetDC(0));
Loader.SetFontProvider(fontProvider);
```

GfX::FontProviderWin32 Windows Display Context
DC (::GetDC(0))

4.4.1

Scaleform

font APIs (Gfx::FontProviderWin32 and

Gfx::FontProviderFT2)

```
Font::NativeHintingRange vectorRange;  
Font::NativeHintingRange rasterRange;  
unsigned maxVectorHintedSize;  
unsigned maxRasterHintedSize;
```

VectorRange RasterRange

Font::DontHint -

Font::HintCJK -

Font::HintAll -

RasterRange VectorRange maxVectorHintedSize maxRasterHintedSize

SimSun

abcdefghijklmnopqrstuvwxyz	vectorRange=GfxFont::DontHint
ABCDEFGHIJKLMNOPQRSTUVWXYZ	rasterRange=GfxFont::DontHint

𐀀𐀁𐀂𐀃𐀄𐀅𐀆𐀇𐀈𐀉𐀊𐀋𐀌𐀍𐀎𐀏𐀐𐀑𐀒𐀓𐀔𐀕𐀖𐀗𐀘𐀙𐀚
𐀛𐀜𐀝𐀞𐀟𐀠𐀡𐀢𐀣𐀤𐀥𐀦𐀧𐀨𐀩𐀪𐀫𐀬𐀭𐀮𐀯𐀰𐀱𐀲𐀳𐀴𐀵𐀶𐀷𐀸𐀹𐀺

abcdefghijklmnopqrstuvwxyz	vectorRange=GfxFont::HintAll
ABCDEFGHIJKLMNOPQRSTUVWXYZ	rasterRange=GfxFont::DontHint

𐀀𐀁𐀂𐀃𐀄𐀅𐀆𐀇𐀈𐀉𐀊𐀋𐀌𐀍𐀎𐀏𐀐𐀑𐀒𐀓𐀔𐀕𐀖𐀗𐀘𐀙𐀚
𐀛𐀜𐀝𐀞𐀟𐀠𐀡𐀢𐀣𐀤𐀥𐀦𐀧𐀨𐀩𐀪𐀫𐀬𐀭𐀮𐀯𐀰𐀱𐀲𐀳𐀴𐀵𐀶𐀷𐀸𐀹𐀺

abcdefghijklmnopqrstuvwxyz	vectorRange=GfxFont::HintAll
ABCDEFGHIJKLMNOPQRSTUVWXYZ	rasterRange=GfxFont::HintCJK

𐀀𐀁𐀂𐀃𐀄𐀅𐀆𐀇𐀈𐀉𐀊𐀋𐀌𐀍𐀎𐀏𐀐𐀑𐀒𐀓𐀔𐀕𐀖𐀗𐀘𐀙𐀚
𐀛𐀜𐀝𐀞𐀟𐀠𐀡𐀢𐀣𐀤𐀥𐀦𐀧𐀨𐀩𐀪𐀫𐀬𐀭𐀮𐀯𐀰𐀱𐀲𐀳𐀴𐀵𐀶𐀷𐀸𐀹𐀺

“Arial Unicode MS”

abcdefghijklmnopqrstuvwxyz
 ABCDEFGHIJKLMNOPQRSTUVWXYZ
 僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞
 僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞

Adobe Flash Scaleform

Win32 FreeType

```

vectorRange = Font::DontHint;
rasterRange = Font::HintCJK;
maxVectorHintedSize = 24;
maxRasterHintedSize = 24;
  
```

4.4.2

SetHinting() SetHintingAllFonts()

```
fontProvider->SetHintingAllFonts(. . .);
```

```
fontProvider->SetHinting(fontName, . . .);
```

SetHinting() SetHintingAllFonts()

```

void SetHintingAllFonts(Font::NativeHintingRange vectorRange,
                        Font::NativeHintingRange rasterRange,
                        unsigned maxVectorHintedSize=24,
                        unsigned maxRasterHintedSize=24);
  
```

```

void SetHinting(const char* Name,
                Font::NativeHintingRange vectorRange,
                Font::NativeHintingRange rasterRange,
                unsigned maxVectorHintedSize=24,
                unsigned maxRasterHintedSize=24);
  
```

Name UTF-8

4.5 Gfx::FontProviderFT2

David Turner FreeType-2 library
 Gfx::FontProviderWin32 *Gfx::FontProviderFT2*

FreeType

Windows MSVC

```

freetype<ver>.lib –          DLL
freetype<ver>_D.lib –        DLL
freetype<ver>MT.lib –        DLL(      CRT)
freetype<ver>MT_D.lib –      DLL (      CRT)

“<ver>”      FreeType          2.1.9      219

freetype2/include  freetype2/objs
  
```

Gfx::FontProviderFT2

```

#include "Gfx/Gfx_FontProviderFT2.h"
. . .
Ptr<FontProviderFT2> fontProvider = *new FontProviderFT2;
<Map Font to Files or Memory>
Loader.SetFontProvider(fontProvider);
  
```

FontProviderFT2(FT_Library lib=0);

```

FreeType                                     FreeType
      FreeType                               Scaleform
      (
FT_Done_FreeType)                           Gfx::Loader
      FreeType
fontProvider->GetFT_Library()                 FT_Library
      FreeType                               Gfx::FontProviderFT2
FreeType
  
```

4.5.1 FreeType

Win32 API

FreeType

Gfx::FontProviderFT2

```
void MapFontToFile(const char* fontName, unsigned fontFlags,
                  const char* fileName, unsigned faceIndex=0,
                  Font::NativeHintingRange vectorHintingRange = Font::DontHint,
                  Font::NativeHintingRange rasterHintingRange = xFont::HintCJK,
                  unsigned maxVectorHintedSize=24,
                  unsigned maxRasterHintedSize=24);
```

“fontName”	“Times New Roman”	“filename”
“C:\\WINDOWS\\Fonts\\times.ttf”	“fontFlags”	“Font::FF_Bold
“Font::FF_Italic	“Font::FF_BoldItalic”	“Font::FF_Bold
Font::FF_Italic”	“faceIndex”	0
	FT_New_Face	;
	Win32	
FreeType		

4.5.2 將字體映射到記憶體

FreeType

```
void MapFontToMemory(const char* fontName, unsigned fontFlags,
                    const char* fontData, unsigned dataSize,
                    unsigned faceIndex=0,
                    Font::NativeHintingRange vectorHintingRange = Font::DontHint,
                    Font::NativeHintingRange rasterHintingRange = Font::HintCJK,
                    unsigned maxVectorHintedSize=24,
                    unsigned maxRasterHintedSize=24);
```

```
FILE* fd = fopen("C:\\WINDOWS\\Fonts\\times.ttf", "rb");
if (fd) {
    fseek(fd, 0, SEEK_END);
    unsigned size = ftell(fd);
    fseek(fd, 0, SEEK_SET);
    char* font = (char*)malloc(size);
    fread(font, size, 1, fd);
    fclose(fd);
}
```

```

    fontProvider->MapFontToMemory("Times New Roman", 0, font, size);
}

```

Gfx::Loader

FreeType

Gfx

Windows

FreeType

```

Ptr<FontProviderFT2> fontProvider = *new FontProviderFT2;
fontProvider->MapFontToFile("Times New Roman", 0,
    "C:\\WINDOWS\\Fonts\\times.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_Bold,
    "C:\\WINDOWS\\Fonts\\timesbd.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_Italic,
    "C:\\WINDOWS\\Fonts\\timesi.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_BoldItalic,
    "C:\\WINDOWS\\Fonts\\timesbi.ttf");

fontProvider->MapFontToFile("Arial", 0,
    "C:\\WINDOWS\\Fonts\\arial.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_Bold,
    "C:\\WINDOWS\\Fonts\\arialbd.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_Italic,
    "C:\\WINDOWS\\Fonts\\ariali.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_BoldItalic,
    "C:\\WINDOWS\\Fonts\\arialbi.ttf");

fontProvider->MapFontToFile("Verdana", 0,
    "C:\\WINDOWS\\Fonts\\verdana.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_Bold,
    "C:\\WINDOWS\\Fonts\\verdanab.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_Italic,
    "C:\\WINDOWS\\Fonts\\verdanai.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_BoldItalic,
    "C:\\WINDOWS\\Fonts\\verdanaz.ttf");

. . .
Loader.SetFontProvider(fontProvider);

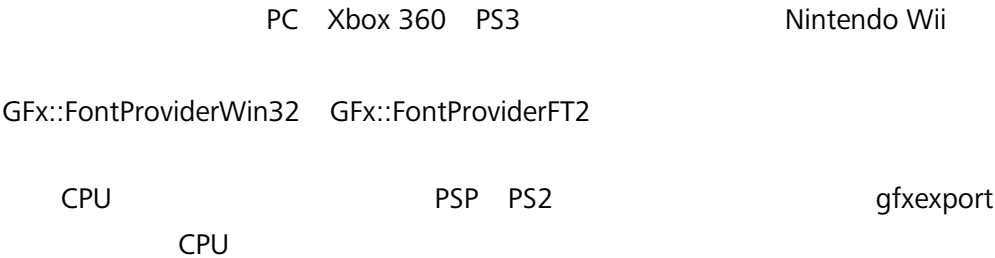
```

MapFontToFile()

5 4

Scaleform

- 1. / gfxexport
- 2.
- 3. Gfx::FontPackParams
- 4.



Scaleform “ ”

5.1 配置字形缓存

Scaleform	Render::GlyphCacheConfig	Gfx::FontPackParams
,	Renderer2D	,
Scaleform		

Gfx 4.0

Scaleform 3.x	4.0	Gfx 3.3	GfxLoader
	GfxFontCacheManager	Gfx 4.0	Renderer2D
GlyphCacheConfig	Scaleform Gfx 4.0		5.2

Gfx::FontPackParams 'gfxexport'

```

bool Done = false;

if (Font has Static Textures with Packed Glyphs)
{
    if (GlyphSize <
        FontPackParams.TextureConfig.NominalSize * MaxRasterScale)
    {
        Draw the Glyph as a Texture using Static Cache;
        Done = true;
    }
}
else if (Dynamic Cache is Enabled AND
        GlyphSize < GlyphCacheParams.MaxSlotHeight)
{
    Draw the Glyph as a Texture using Dynamic Cache;
    Done = True;
}

if (Not Done)
{
    Draw the Glyph as Vector Shape;

```

```
}
```

5.2

10000

```
renderer->GetGlyphCacheConfig()->SetParams(Render::GlyphCacheParams(0));
```

```
,renderer Render::Renderer2D ,
```

HAL

LRU

```
Render::GlyphCacheParams gcparams;  
gcparams.TextureWidth = 1024;  
gcparams.TextureHeight = 1024;  
gcparams.MaxNumTextures = 1;  
gcparams.MaxSlotHeight = 48;  
gcparams.SlotPadding = 2;  
gcparams.TexUpdWidth = 256;  
gcparams.TexUpdHeight = 512;
```

```
renderer->GetGlyphCacheConfig()->SetParams(gcparams);
```

TextureWidth, TextureHeight -

2

MaxNumTextures -

MaxSlotHeight -

SlotPadding -

TexUpdWidth, TexUpdWidth - 256x256 128x128 256x512 (128K)

4KB 4K 16 100 256 1MB 4 KB 256 256 4K 1MB

MaxSlotHeight+2*SlotPadding 2-5 10 60-80%

70-75 500 2000

“ ” “ ”

Flash

GFX, Dynamic cache	Readability	Animation
<i>Anti-alias for readability</i> <i>Anti-alias for animation</i>		

“ ” Scaleform
Flash Scaleform

(O G C Q o g e)
Scaleform

- H, E, F, T, U, V, W, X, Z
- z, x, v, w, y

Scaleform

“Warning: Font 'Arial': No hinting chars (any of 'HEFTUVWXZ' and 'zxvwy'). Auto-Hinting is disabled.”

'Arial': ('HEFTUVWXZ' and 'zxvwy')

Flash “Zz” “Basic Latin” “ ”

5.3 – *gfxexport*

gfxexport SWF , GFX , ,DDS
SWF ,
TGA -fc ,gfxexport , ,

-fc	
-fcl <size>	256 , 256 25% (Flash 1024), ,
-fcm	256 , 256 25% (Flash 1024), ,

5.4 - *gfxexport*

gfxexport

Scaleform Player

Scaleform

GFx

gfxexport

-
-
-

CPU

Gfx::ImageCreator

Render::Renderer

“ ”

'test.swf'

'test.gfx'

```
gfxexport -fonts -strip_font_shapes test.swf
```

`-strip_font_shapes`

Scaleform

gfxexport

gfxexport

-fonts	
-fns <size>	48 tri-linear mip-map
-fpp <n>	3
-fts <WxH>	256x256 '-fts 128' 128x128' -fts 512x128'


```

SetTextureConfig()
    FontPackParams::TextureConfig fontPackConfig;
    fontPackConfig.NominalSize    = 48;
    fontPackConfig.PadPixels      = 3;
    fontPackConfig.TextureWidth   = 1024;
    fontPackConfig.TextureHeight  = 1024;

```

NominalSize -

```

    “NominalSize”
                                NominalSize
                                “      ”

```

PadPixels — 1

TextureWidth, TextureHeight - 2

- 1) ;
- 2)

```
Ptr<FontPackParams> packParams = *new FontPackParams();
Loader.SetFontPackParams(packParams);
...
renderer->GetGlyphCacheConfig()->SetParams(Render::GlyphCacheParams(0));
```
- 3)

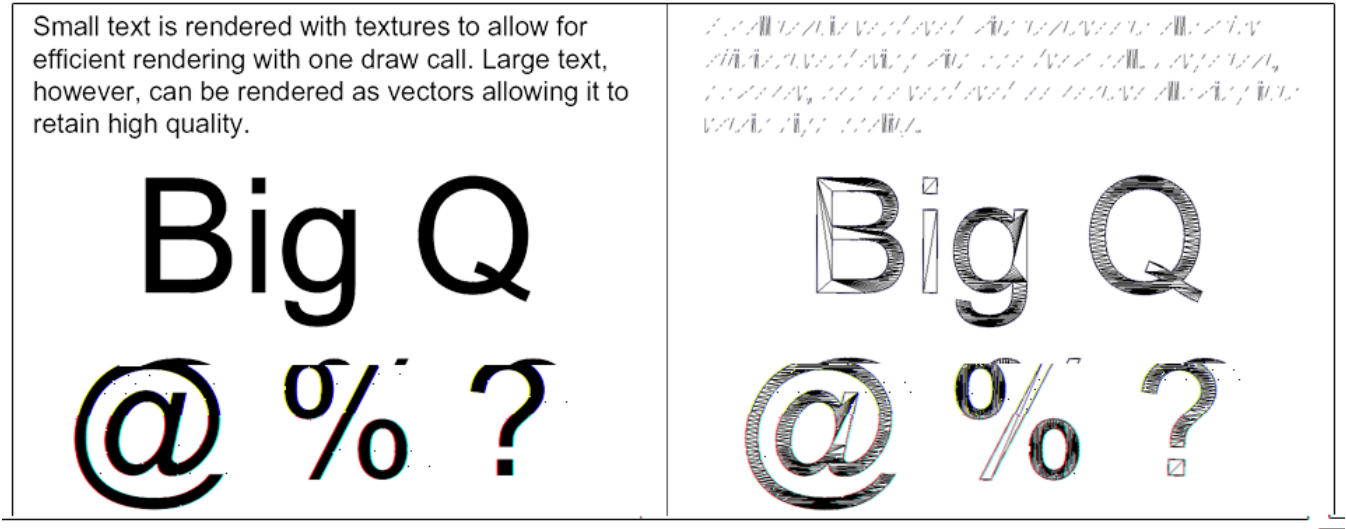
```
Ptr<FontPackParams> packParams = *new FontPackParams();
Loader.SetFontPackParams(packParams);
Loader.GetFontPackParams()->SetGlyphCountLimit(500);
500
```

5.6

Flash
Scaleform

Scaleform

Scaleform Player
Ctrl+W



Render::GlyphCacheConfig::SetParams
MaxSlotHeight

GlyphCacheParams::MaxRasterScale

SetMaxRasterScale
MaxRasterScale 1 1.25
1.25
60

48

MaxRasterScale
gfxexport

Scaleform

-strip_font_shapes

6 5

Scaleform 2.x

Scaleform

Flash

Adobe Flash ,

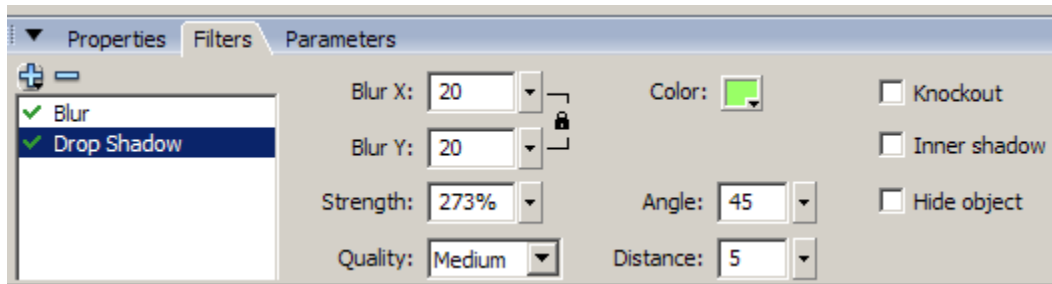
Flash

Scaleform

Scaleform

6.1

Flash Studio



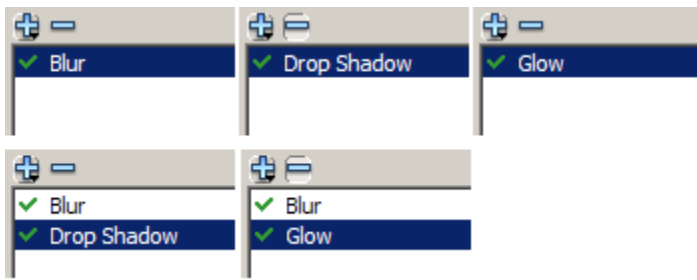
Flash Scaleform
Flash

Scaleform

Scaleform
Flash

Scaleform

-
- “ ” “ ”
-
-
-



Scaleform Flash “ ” “ ” “ ” “ ”

“Knockout” “Hide Object”

Scaleform “Inner Shadow” “Inner Glow”

15.9 Scaleform X Y

1590

“Knockout” Flash

Scaleform

Flash GfX

Text *Text*

“Knockout”

6.2

Flash 3x3 3x3

;

Flash

Scaleform

Scaleform

“Quality: Low”

Scaleform
(sigma)

Scaleform

	Low quality	Medium quality	High quality
Flash	<i>Text</i>	<i>Text</i>	<i>Text</i>
GFx	<i>Text</i>	<i>Text</i>	<i>Text</i>

Flash

; Scaleform

Scaleform

<http://www.ph.tn.tudelft.nl/Courses/FIP/noframes/fip-Smoothin.html>.

6.3

Flash

Scaleform

Scaleform

/

6.4 *ActionScript*

Scaleform (DropShadowFilter BlurFilter ColorMatrixFilter BevelFilter),
AS2 AS3 / , Flash